DP-OPS-0017: Context Bundle Generator & Tiered LLM Artifacts

0. FRESHNESS GATE (MUST PASS BEFORE WORK)

Base Branch: main

Required Work Branch: work/ops-llms-gen-2026-02-03

Base HEAD: 9ab382c78

Required local re-check (worker runs; paste results in RESULTS):

- git rev-parse --abbrev-ref HEAD
- git rev-parse HEAD
- git status --porcelain

Worker must already be on the Required Work Branch listed above (no branch creation or switching). STOP if any mismatch. STOP if the Required Work Branch is missing. STOP if told to create or switch branches.

I) REQUIRED CONTEXT LOAD (READ BEFORE DOING ANYTHING)

Worker must confirm loaded before acting:

- TRUTH, AGENTS, SoP, CONTEXT, MAP
- Plus any DP-scoped files listed here:
  - ops/lib/skill/skill_lib.sh (reference for redaction logic)
  - tools/lint_truth.sh (reference for file lists)

(OPEN is for Planner/DP Writer state refresh; Worker does not read OPEN). DISPOSABLE ARTIFACTS (chat logs, etc.) MUST NOT BE REFERENCED OR INCLUDED.

II) SCOPE & SAFETY

- **Objective:** Implement ops/bin/llms-gen to deterministically generate llms-small.txt and

llms-full.txt (tiered context bundles) and enforce them via tools/lint_llms.sh, preventing drift and ensuring high-signal context for AI ingestion.
- **Non-Goals (optional, drift-killer):**
  - Do not modify existing dump logic (ops/bin/dump).
  - Do not create "medium" or custom tiers; strictly Small and Full.

Target Files allowlist (hard gate)

- (new) ops/bin/llms-gen
- (new) tools/lint_llms.sh
- tools/lint_truth.sh
- llms.txt
- ops/lib/manifests/CONTEXT.md
- SoP.md
- SKILL.md
- docs/library/INDEX.md
- (new) docs/library/skills/S-LEARN-08.md

Allowlist rule: exact paths only; use (new) prefix only when the DP explicitly allows new files.

- **Conditional Requirements:** If canon/governance surfaces (e.g., TASK.md) are modified, SoP.md MUST be included in the allowlist and updated.
- **Stop Condition (scope):** If any required change falls outside the allowlist, **STOP** and report.
- **Forbidden (unless DP explicitly overrides):**
  - No edits to .github/
  - No edits to projects/ unless explicitly authorized
  - No renames/moves/deletes unless explicitly authorized
- **Worker Git Authority (non-negotiable):**
  - Workers do not create/switch branches.
  - Workers do not stage/commit/merge/push.
  - Workers only return diffs/results/receipt.
- **Precedence:** If any conflict exists, **TRUTH.md wins**. If unclear, STOP and ask.
- **Strict Stop Conditions (generic):**
  - STOP if any placeholder text remains in final outputs ([...], TODO, TBD, ...).
  - STOP if required DP inputs are missing (do not guess).
  - STOP if IDs / naming constraints in the DP are inconsistent (when applicable).
- **Constraint Clarification (Placeholder Exemption):**
  - The generic Stop Condition regarding placeholders (TODO, ...) applies to **newly authored logic and documentation**.
  - It is **explicitly waived** for the generated artifacts (llms-small.txt, llms-full.txt) where such tokens exist verbatim in the source text.
  - **Directive:** Do not filter or redact source tokens from the generated bundles. Maintain 100% fidelity to the source code state, as these artifacts describe the repo "as is."

III. EXECUTION PLAN (A–E CANON)

A) STATE

- **Context:** Currently, providing context to LLMs requires manual dump execution or pointing to disparate files (TRUTH.md, AGENTS.md). There is no persistent, machine-readable "high-signal" bundle at the repo root.
- **Drift:** llms.txt points to files, but does not inline them, requiring multiple hops for retrieval. Context bundles (if manually created) drift from the codebase immediately.
- **Desired State:** Two deterministic artifacts (llms-small.txt, llms-full.txt) exist at the repo root, generated by a script (ops/bin/llms-gen) and policed by a linter (tools/lint_llms.sh). These artifacts are typically high-signal (Small) or comprehensive (Full) and are strictly read-only outputs of the generator.

B) REQUEST

Numbered tasks (concrete, no vibes):

1. **Create ops/bin/llms-gen (Bash, executable):**
   - **Inputs:** git ls-files (source of truth).
   - **Tiers:**
     - small: Includes TRUTH.md, AGENTS.md, docs/library/INDEX.md, and the first 10 skills (docs/library/skills/S-LEARN-0[1-9].md + 10.md).
     - full: Includes small set + README.md (summary only), all files in docs/library/ (recursive), and a "freshness slice" of SoP.md (head 10 entries).
   - **Processing:**
     - Iterate files in deterministic order (sorted).
     - Prepend ## <repo-relative-path> header to each file content.
     - Strip "Table of Contents" blocks (lines containing `` or looking like * [Link](#...)) to save tokens.
     - **Redaction:** Apply regex patterns to scrub secrets (API keys, private keys) using logic mirrored from ops/lib/skill/skill_lib.sh (e.g., sed replacements for known patterns).
   - **Output:** Write to llms-small.txt and llms-full.txt in the repo root.
2. **Create tools/lint_llms.sh (Bash, executable):**
   - **Logic:** Run ops/bin/llms-gen to a temporary location.
   - **Check:** Diff generated files against the actual llms-small.txt and llms-full.txt in the root.
   - **Failure:** If diff exists, print "FAIL: LLM artifacts drifted. Run ops/bin/llms-gen." and exit 1.
   - **Success:** Print "OK: LLM artifacts fresh."
3. **Update tools/lint_truth.sh:**

    - Add llms-small.txt and llms-full.txt to the root_files array to ensure they are monitored for spelling/naming validity (e.g., ensuring "Stela" is spelled correctly within the bundle headers).
4. **Update llms.txt:**
    - Update content to explicitly reference llms-small.txt (Core Context) and llms-full.txt (Extended Context) as primary entry points.
    - Clarify that llms.txt is the pointer, while llms-*.txt are the payloads.
5. **Update ops/lib/manifests/CONTEXT.md:**
    - Add a note in "Verification" or "Canonical Context" referencing llms-small.txt as a valid quick-load artifact for agents.
6. **Harvest Skill (S-LEARN-08):**
    - **Task:** Capture the ops/bin/llms-gen workflow as a skill.
    - **Name:** "LLM Context Bundling".
    - **Context:** When context artifacts are missing or stale.
    - **Solution:** Run ops/bin/llms-gen to regenerate headers and content; verify with tools/lint_llms.sh.
    - **Action:** Harvest, Refine, and Promote to docs/library/skills/S-LEARN-08.md.
7. **Update SoP.md:**
    - Log the creation of the generator, the new canon artifacts, and the exemption decision regarding placeholders.

C) CHANGELOG

- **Log (human-readable, 1–6 bullets):**
    - Created ops/bin/llms-gen to standardize LLM context bundling (Small/Full tiers).
    - Created tools/lint_llms.sh to enforce bundle freshness in CI.
    - Updated llms.txt and ops/lib/manifests/CONTEXT.md to point to new artifacts.
    - Updated tools/lint_truth.sh to police new root files.
    - Harvested and promoted S-LEARN-08 (LLM Context Bundling).

D) PATCH / DIFF

- **Format:** Unified diff (preferred) or anchored snippets.

E) RECEIPT (REQUIRED)

Verification (MUST RUN; or report NOT RUN + reason + risk)

- ./ops/bin/dump --scope=platform (or repo-correct equivalent when scope differs)
- If dump refiners are used, include this fallback line in the DP:
    - Fallback:./ops/bin/dump --scope=platform --format=chatgpt --out=auto --bundle
- Context lint (use repo-canonical command):
    - Prefer: bash tools/context_lint.sh
    - If not found and required: **STOP** and ask (do not invent a substitute)
- Truth lint (when TRUTH/canon surfaces touched; use repo-canonical command):

    - Prefer: bash tools/lint_truth.sh
    - If not found and required: **STOP** and ask (do not invent a substitute)

Skill capture (required during normal DP processing)

- Workflow: run ops/lib/skill/skill_lib.sh harvest to auto-generate a draft with provenance and semantic collision checks.
- Review: inspect the draft in storage/handoff/ for heuristic accuracy (Hot Zone and High Churn). Refine Invocation guidance and Solution. Do not edit the Provenance block.
- Promote: run ops/lib/skill/skill_lib.sh promote <draft_path> or log "no new skill promoted" with a rationale in RESULTS.
- Legacy capture is forbidden. Do not manually write skill files; the provenance block is required.
- If a DP requires skill capture, the DP Target Files allowlist must include SKILL.md.
- Proof required in RESULTS:
    - Harvest-only: command invoked plus draft path and ls storage/handoff/ output.
- Promotion: command invoked plus grep -n "Promotion Packet:" SKILL.md and grep -n "docs/library/skills/S-LEARN-" docs/library/INDEX.md.

Artifact bundle (required directory)

- storage/handoff/

Proof bundle checklist (required; naming pattern over exact names)

- storage/handoff/DP-OPS-0017-RESULTS.md (required)
- storage/handoff/OPEN-PORCELAIN-work-ops-llms-gen-2026-02-03-[hash].txt (required)
- storage/handoff/OPEN-work-ops-llms-gen-2026-02-03-[hash].txt (required)
- storage/dumps/dump-full-... (or platform scope as applicable)
- git diff --name-only (include in RESULTS)
- git diff --stat (include in RESULTS)
- NEXT: one single action (include in RESULTS)

MANDATE: No receipt/proof bundle = **AUTOMATIC DISAPPROVAL**.

Closeout routine receipts not being produced CONSISTENTLY INCLUDING PORCELAIN, DUMP AND OPEN is a failure condition.

Partial receipts (e.g., missing OPEN artifacts when a session was started) are considered failures.

REQUIRED OUTPUT

MUST include a "Status" block at the top of RESULTS:

- Scope summary: (1 sentence validating the loop)

- Tracked change: (What was edited)
- Verification: (Output of proofs)

RESULTS file must include (minimum)

- Verification results (RUN/NOT RUN + reason + risk)
- Patch/diff proof bundle (status/diff list or direct diff)
- Receipt pointers (OPEN + PORCELAIN + DUMP outputs/paths)
- git diff --name-only output
- git diff --stat output
- NEXT: one single action

Storage artifact flow (local-only, untracked):

- OPEN and OPEN-PORCELAIN outputs land in storage/handoff/.
- Dump bundles land in storage/dumps/.
- Results receipts land in storage/handoff/DP-OPS-0000-RESULTS.md (pattern).
- Operator stores DP drafts under storage/dp/intake/ and moves or copies to storage/dp/processed/ after completion (or when reusing a previously executed DP).
- These are retention paths only; they are not worker prerequisites unless the DP explicitly says so.

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3) CLOSEOUT (MANDATORY)

Worker closeout duties (end of session)

- [ ] Produce/Update: storage/handoff/DP-OPS-0017-RESULTS.md
- [ ] If canon/governance surfaces were changed, include **SoP.md** in scope and update it in the same PR slice.
- [ ] Ensure proof bundle checklist is satisfied (above). (No bundle = DISAPPROVE.)
- [ ] Fill out the Mandatory Closing Block (varied wording required).
- [ ] Append a **TASK Work Log** entry (below) capturing:
  - what changed (high level)
  - gates/verification outcome (pass/fail or not run + reason)
  - blockers (if any)
  - **NEXT:** one single action

Mandatory Closing Block (Varied Wording Required)

Varied Wording provision: Each entry must use meaningfully distinct wording; copy or minor tense changes are not acceptable. Entry 4 must differ from Entry 1.

1. Primary Commit Header (plaintext)
2. Pull Request Title (plaintext)
3. Pull Request Description (markdown)

4. Final Squash Stub (plaintext) (Must differ from #1)
5. Extended Technical Manifest (plaintext)
6. Review Conversation Starter (markdown)

Integrator/operator closeout duties

- [ ] Review diff vs DP scope/forbidden zones
- [ ] Run repo-gates / required checks
- [ ] Commit/push/open PR/merge (operator-controlled)

3.1) THREAD TRANSITION (RESET / ARCHIVE RULE)

Purpose: Prevent TASK from becoming a museum while preserving continuity.

When a DP is **complete** (merged) or **ended** (canceled/superseded):

1. Add a final Work Log entry that starts with: THREAD END: and includes:
   - outcome (merged/canceled/superseded)
   - pointer to the DP RESULTS file
   - NEXT: the single action (often “start new DP” or “await operator”)
2. Reset sections **1) ACTIVE CONTEXT** and **2) CURRENT DISPATCH PACKET** to the next DP.
3. Keep the Work Log, but separate threads with a visible divider line:
   - ---
   - THREAD START: DP-OPS-

(If the Work Log gets too long, it’s acceptable to keep only the most recent thread’s entries here and move older thread logs into the DP RESULTS file—only when explicitly directed.)

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4) WORK LOG (TIMESTAMPED CONTINUITY)

Rule: This is not a transcript.

It’s the durable breadcrumb trail.

Expectation: Append a new timestamped entry on every DP closeout (no exceptions).

Each entry ends with: NEXT: <one single action>.

- **Example format:** 2026-01-27 14:05 — DP-OPS-0005: Example entry only. Verification: NOT RUN. Blockers: none. NEXT: follow-up.
- ** — DP-OPS-: [short note]. Verification:. Blockers: [none/...]. NEXT: [single action]
- ** — [short note]. Verification: [..]. Blockers: [..]. NEXT: [single action]